TASK 2:Exploratory Data Analysis(EDA

❖ INTRODUCTION:

This dataset contains information about students including their names,mobile number,grades,marks,comments,roll no ,school name ,address.The purpose of this analysis is to understand patterns and relationships within the data .

❖ DATASET DESCRIPTION:

- Total number of rows:500
- Total number of columns:10
- Column names: student_names,phone_no,maths,physics ,chemistry ,grade, comment ,roll no,school name ,student address

❖ DATA TYPES:

- Name: categorical data
- Phone number:categorical data
- Subjects:categorical data
- Grade:categorical data
- Comment:categorical data
- Roll no: numerical data(identifier)
- School name:categorical data
- Student address: categorical data

❖ STATISTICAL ANALYSIS:

Since the dataset mainly contains categorical data ,statistical analysis is performed using frequency counts.

- Number of students with grade A:21
- Number of students with grade A+:2
- Number of students with grade B:94
- Number of students with grade B+:53
- Number of students with grade C:125
- Number of students with grafe D:169
- Most common grade:D

As the dataset contains mostly categorical variables ,measures like mean and median are not applicable

❖ **Observations:**

- The majority of students fall under grade D ,indicating that a large number of students have lower academic performance
- Grade C is the second most common grade ,showing that many students have average performance
- Grade B also has a significamt number of students ,indicating moderate performance levels
- Very few students achieved Grade A +,which shows that high academic performance is limited
- Overall the distribution of grades suggests that most students are performing at an average to performing at an average to below-average level.

❖ **DATA ISSUES:**

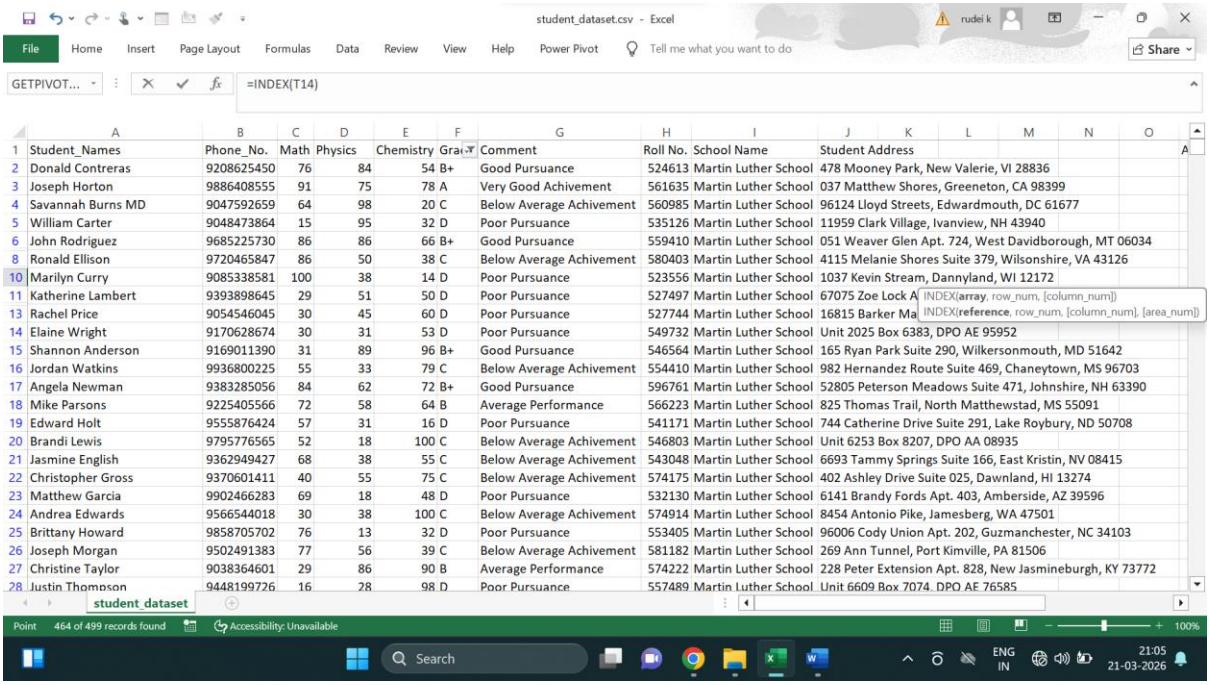
- No major missing values were found in dataset
- Most of data is categorical .so numerical statistical methods like mean ,median, are not applicable.

- Roll numbers and phone numbers are numerical in form but are used as identifiers ,so they were treated as categorical data.

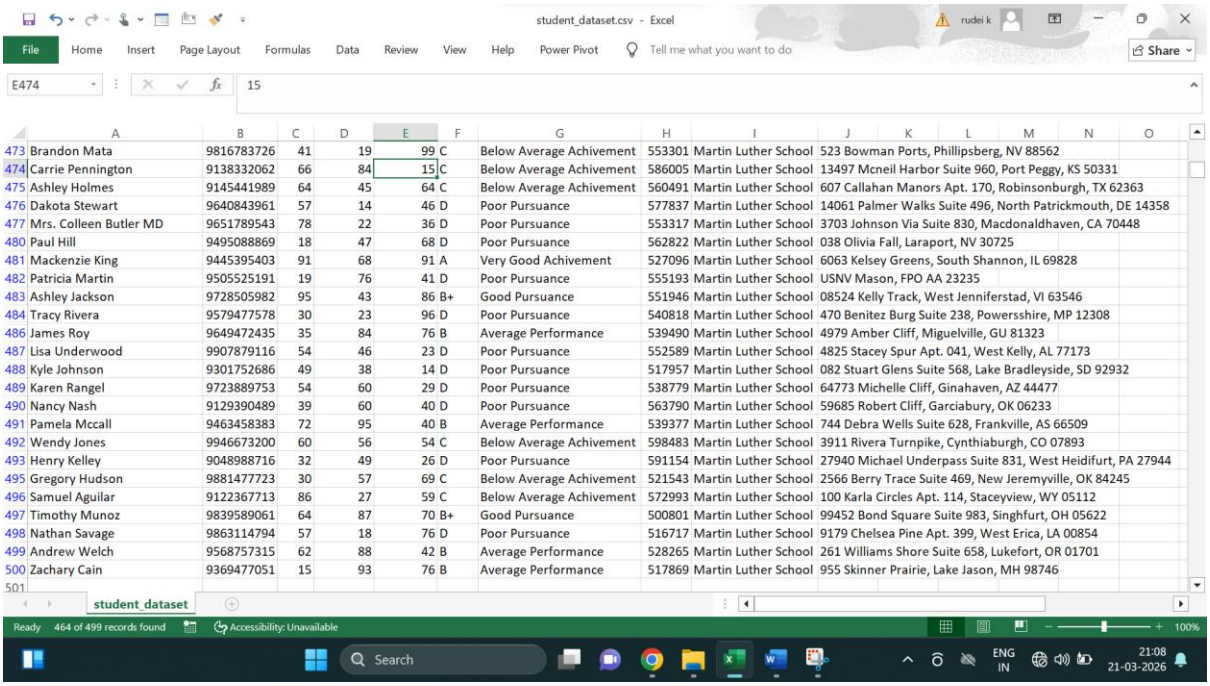
❖ CONCLUSION:

The analysis of the dataset shows that the majority of students fall under grade D,indicating that overall academic performance is on the lower side. A significant number of students are in grade B and C ,representing average performance levels.very few students achieved grade A ,showing limited high performance.
Overall the performance suggest that there is a need for improvement in student performance ,and more focus should be given to help students achieve higher grades.

❖ DATASET:



Student_Names	Phone_No.	Math	Physics	Chemistry	Grade	Comment	Roll No.	School Name	Student Address
Donald Contreras	9208625450	76	84	54	B+	Good Pursuance	524613	Martin Luther School	478 Mooney Park, New Valerie, VI 28836
Joseph Horton	9886408555	91	75	78	A	Very Good Achivement	561635	Martin Luther School	037 Matthew Shores, Greeneton, CA 98399
Savannah Burns MD	9047592659	64	98	20	C	Below Average Achivement	560985	Martin Luther School	96124 Lloyd Streets, Edwardmouth, DC 61677
William Carter	9048473864	15	95	32	D	Poor Pursuance	535126	Martin Luther School	11959 Clark Village, Ivanview, NH 43940
John Rodriguez	9685225730	86	86	66	B+	Good Pursuance	559410	Martin Luther School	051 Weaver Glen Apt. 724, West Davidborough, MT 06034
Ronald Ellison	9720465847	86	50	38	C	Below Average Achivement	580403	Martin Luther School	4115 Melanie Shores Suite 379, Wilsonshire, VA 43126
Marilyn Curry	9085338581	100	38	14	D	Poor Pursuance	523556	Martin Luther School	1037 Kevin Stream, Dannyland, WI 12172
Katherine Lambert	9393898645	29	51	50	D	Poor Pursuance	527497	Martin Luther School	67075 Zoe Lock A
Rachel Price	9054546045	30	45	60	D	Poor Pursuance	527744	Martin Luther School	16815 Barker Ma
Elaine Wright	9170628674	30	31	53	D	Poor Pursuance	549732	Martin Luther School	Unit 2025 Box 6383, DPO AE 95952
Shannon Anderson	9169011390	31	89	96	B+	Good Pursuance	546564	Martin Luther School	165 Ryan Park Suite 290, Wilkersonmouth, MD 51642
Jordan Watkins	9936800225	55	33	79	C	Below Average Achivement	554410	Martin Luther School	982 Hernandez Route Suite 469, Chaneytown, MS 96703
Angela Newman	9383285056	84	62	72	B+	Good Pursuance	596761	Martin Luther School	52805 Peterson Meadows Suite 471, Johnshire, NH 63390
Mike Parsons	9225405566	72	58	64	B	Average Performance	566223	Martin Luther School	825 Thomas Trail, North Matthewstad, MS 55091
Edward Holt	9555876424	57	31	16	D	Poor Pursuance	541171	Martin Luther School	744 Catherine Drive Suite 291, Lake Roybury, ND 50708
Brandi Lewis	9795776565	52	18	100	C	Below Average Achivement	546803	Martin Luther School	Unit 6253 Box 8207, DPO AA 08935
Jasmine English	9362949427	68	38	55	C	Below Average Achivement	543048	Martin Luther School	6693 Tammy Springs Suite 166, East Kristin, NV 08415
Christopher Gross	9370601411	40	55	75	C	Below Average Achivement	574175	Martin Luther School	402 Ashley Drive Suite 025, Dawnland, HI 13274
Matthew Garcia	9902466283	69	18	48	D	Poor Pursuance	532130	Martin Luther School	6141 Brandy Fords Apt. 403, Amberside, AZ 39596
Andrea Edwards	9566544018	30	38	100	C	Below Average Achivement	574914	Martin Luther School	8454 Antonio Pike, Jamesberg, WA 47501
Brittany Howard	9858705702	76	13	32	D	Poor Pursuance	553405	Martin Luther School	96006 Cody Union Apt. 202, Guzmanchester, NC 34103
Joseph Morgan	9502491383	77	56	39	C	Below Average Achivement	581182	Martin Luther School	269 Ann Tunnel, Port Kimville, PA 81506
Christine Taylor	9038364601	29	86	90	B	Average Performance	574222	Martin Luther School	228 Peter Extension Apt. 828, New Jasmineburgh, KY 73772
Justin Thompson	9448199726	16	28	98	D	Poor Pursuance	557489	Martin Luther School	Unit 6609 Box 7074, DPO AE 76585



473	Brandon Mata	9816783726	41	19	99	C	Below Average Achivement	553301	Martin Luther School	523 Bowman Ports, Phillipsberg, NV 88562
474	Carrie Pennington	9138332062	66	84	15	C	Below Average Achivement	586005	Martin Luther School	13497 Mcneil Harbor Suite 960, Port Peggy, KS 50331
475	Ashley Holmes	9145441989	64	45	64	C	Below Average Achivement	560491	Martin Luther School	607 Callahan Manors Apt. 170, Robinsonburgh, TX 62363
476	Dakota Stewart	9640843961	57	14	46	D	Poor Pursuance	577837	Martin Luther School	14061 Palmer Walks Suite 496, North Patrickmouth, DE 14358
477	Mrs. Colleen Butler MD	9651789543	78	22	36	D	Poor Pursuance	553317	Martin Luther School	3703 Johnson Via Suite 830, Macdonaldhaven, CA 70448
480	Paul Hill	9495088869	18	47	68	D	Poor Pursuance	562822	Martin Luther School	038 Olivia Fall, Laraport, NV 30725
481	Mackenzie King	9445395403	91	68	91	A	Very Good Achivement	527096	Martin Luther School	6063 Kelsey Greens, South Shannon, IL 69828
482	Patricia Martin	9505525191	19	76	41	D	Poor Pursuance	555193	Martin Luther School	USNV Mason, FPO AA 23235
483	Ashley Jackson	9728505982	95	43	86	B+	Good Pursuance	551946	Martin Luther School	08524 Kelly Track, West Jenniferstad, VI 63546
484	Tracy Rivera	9579477578	30	23	96	D	Poor Pursuance	540818	Martin Luther School	470 Benitez Burg Suite 238, Powersshire, MP 12308
486	James Roy	9649472435	35	84	76	B	Average Performance	539490	Martin Luther School	4979 Amber Cliff, Miguelville, GU 81323
487	Lisa Underwood	9907879116	54	46	23	D	Poor Pursuance	552589	Martin Luther School	4825 Stacey Spur Apt. 041, West Kelly, AL 77173
488	Kyle Johnson	9301752686	49	38	14	D	Poor Pursuance	517957	Martin Luther School	082 Stuart Glens Suite 568, Lake Bradleyside, SD 92932
489	Karen Rangel	9723889753	54	60	29	D	Poor Pursuance	538779	Martin Luther School	64773 Michelle Cliff, Ginahaven, AZ 44477
490	Nancy Nash	9129390489	39	60	40	D	Poor Pursuance	563790	Martin Luther School	59685 Robert Cliff, Garciabury, OK 06233
491	Pamela Mccall	9463458383	72	95	40	B	Average Performance	539377	Martin Luther School	744 Debra Wells Suite 628, Frankville, AS 66509
492	Wendy Jones	9946673200	60	56	54	C	Below Average Achivement	598483	Martin Luther School	3911 Rivera Turnpike, Cynthiaburgh, CO 07893
493	Henry Kelley	9048988716	32	49	26	D	Poor Pursuance	591154	Martin Luther School	27940 Michael Underpass Suite 831, West Heidifurt, PA 27944
495	Gregory Hudson	9881477723	30	57	69	C	Below Average Achivement	521543	Martin Luther School	2566 Berry Trace Suite 469, New Jeremyville, OK 84245
496	Samuel Aguilar	9122367713	86	27	59	C	Below Average Achivement	572993	Martin Luther School	100 Karla Circles Apt. 114, Staceyview, WY 05112
497	Timothy Munoz	9839589061	64	87	70	B+	Good Pursuance	500801	Martin Luther School	99452 Bond Square Suite 983, Singhfurt, OH 05622
498	Nathan Savage	9863114794	57	18	76	D	Poor Pursuance	516717	Martin Luther School	9179 Chelsea Pine Apt. 399, West Erica, LA 00854
499	Andrew Welch	9568757315	62	88	42	B	Average Performance	528265	Martin Luther School	261 Williams Shore Suite 658, Lukefort, OR 01701
500	Zachary Cain	9369477051	15	93	76	B	Average Performance	517869	Martin Luther School	955 Skinner Prairie, Lake Jason, MH 98746

TASK 3:DATA VISUALIZATION

❖ INTRODUCTION:

Data visualization is the process of representing data in graphical form such as charts and graphs .It helps in understanding patterns ,trends and insights in an easy and clear way.

❖ DATASET DESCRIPTION:

The dataset contains information about students including their grades and other details .The total number of records in the dataset is 500.

❖ DATA PREPARATION:

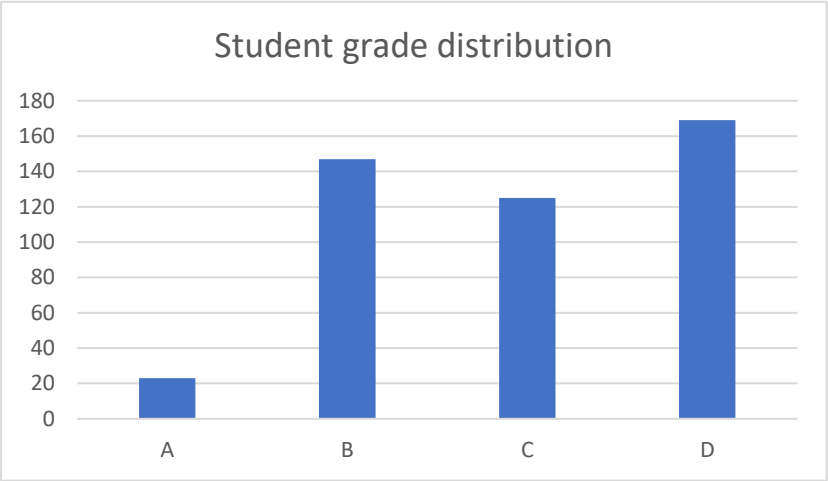
For visualization ,grade data was grouped and counted

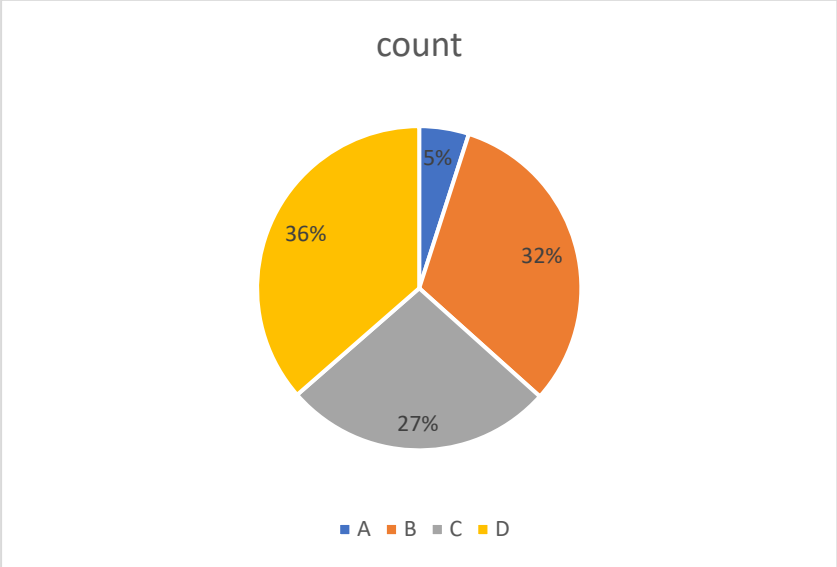
- Grade A=21
- Grade A+=2

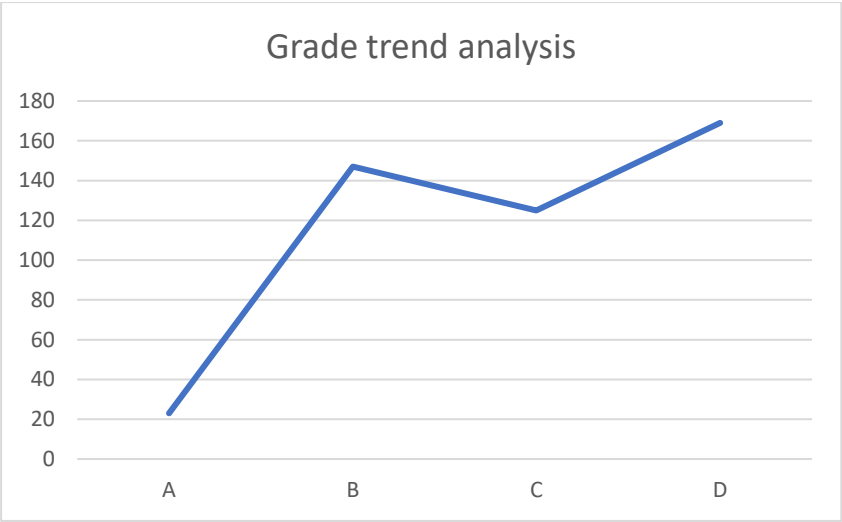
- Grade B=94
- Grade B+=53
- Grade C=125
- Grade D=169

Grades such as A+ and B+ were combined with A and

❖ **VISUALIZATION:**







- **CONCLUSION:** The data visulization of student grades clearly shows the overall performance distribution .It is observed that the majority of student fall in grade D and Grade B,Overall data visualization makes it easier to analyze patterns,compare values and draw meaningful insights for better decision making.